

LAPORAN PRAKTIKUM
MACHINE LEARNING (PRAKTIKUM)



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Tujuan

Menganalisis performa algoritma Support Vector Machine (SVM) dalam mengklasifikasikan data Sleep Health and Lifestyle, serta membandingkan hasil tiga kernel: Linear, Polynomial, dan RBF, menggunakan metrik Akurasi, Presisi, Recall, dan F1-Score.

Kode Program

```
from sklearn.svm import SVC
from sklearn.metrics import confusion_matrix, accuracy_score, precision_score, recall_score, f1_s
```

Memuat semua library yang dibutuhkan untuk membuat model SVM, melakukan scaling data, encoding label, serta menghitung metrik evaluasi (Akurasi, Presisi, Recall, F1-Score).

```
@app.route('/upload', methods=['POST'])
def upload_file():
```

Menerima file CSV dari user, lalu membaca dataset menggunakan pandas. Setelah itu, menampilkan preview tabel dan mendeteksi kolom yang bisa dijadikan target klasifikasi

```
@app.route('/train', methods=['POST'])
def train_model():
```

Bagian inti program. Langkah-langkahnya:

1. Baca dataset dan tentukan kolom target.
2. Pisahkan data fitur (X) dan label (y).
3. Label Encoding untuk kolom kategorikal agar bisa diproses oleh model.
4. Split data menjadi train dan test dengan train_test_split().
5. Standarisasi data menggunakan StandardScaler().
6. Latih tiga model SVM:

```
kernels = {
    'linear': {'kernel': 'linear'},
    'polynomial': {'kernel': 'poly', 'degree': 3},
    'rbf': {'kernel': 'rbf', 'gamma': 'scale'}
}
```

7. Setelah training, setiap model menghasilkan:
 - Confusion matrix

- Akurasi, Presisi, Recall, dan F1-Score
- Visualisasi confusion matrix (dalam format gambar base64)

Perhitungan metrik dilakukan seperti ini:

```
acc = accuracy_score(y_test, y_pred)
prec = precision_score(y_test, y_pred, average=avg)
rec = recall_score(y_test, y_pred, average=avg)
f1 = f1_score(y_test, y_pred, average=avg)
```

Hasilnya disimpan ke dictionary results[kname], lalu ditampilkan di UI dengan grafik heatmap.

```
def plot_confusion_matrix(cm, title):
    sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
```

Membuat heatmap confusion matrix menggunakan Seaborn, yang nantinya ditampilkan di halaman hasil untuk tiap kernel.

```
@app.route('/predict_manual', methods=['POST'])
def predict_manual():
```

Bagian ini memungkinkan pengguna memasukkan nilai fitur secara manual dengan memilih salah satu kernel (Linear / Polynomial / RBF).

Dataset

Dataset yang digunakan: Sleep_health_and_lifestyle_dataset.csv

NO.	Kolom	Deskripsi
1.	Person ID	ID unik tiap individu
2.	Gender	Jenis Kelamin
3.	Age	Usia
4.	Occupation	Pekerjaan
5.	Sleep Duration	Lama tidur (jam)
6.	Quality of Sleep	Skor kualitas tidur
7.	Physical Activity Level	Aktivitas fisik harian
8.	Stress Level	Tingkat stress
9.	BMI Category	Kategori indeks massa tubuh
10.	Blood Pressure	Tekanan darah
11.	Heart Rate	Detak jantung per menit
12.	Daily Steps	Jumlah langkah harian
13.	Sleep Disorder	Label target (Insomnia, Sleep Apnea, atau Normal)

Pembagian Data

Dataset terdiri dari 155 data, dengan pembagian:

Keterangan	Jumlah
Data Latih (Train)	124
Data Uji (Test)	31
Jumlah Kelas	2 (Sleep Disorder / Normal)

Metode

Algoritma SVM (Support Vector Machine) digunakan untuk klasifikasi dengan tiga kernel:

- Linear
- Polynomial
- RBF

Evaluasi dilakukan menggunakan Confusion Matrix, untuk menghitung:

- Accuracy
- Precision
- Recall
- F1-Score

Hasil dan Perbandingan

a. Linear Kernel

Hasil Evaluasi:

Metrik	Nilai
Akurasi	83.87%
Presisi	86.67%
Recall	81.25%
F1-Score	83.87%

Confusion Matrix	Prediksi: Disorder	Prediksi: Normal
Aktual: Disorder	13	3
Aktual: Normal	2	13

b. Polynomial Kernel

Hasil Evaluasi:

Metrik	Nilai
Akurasi	87.10%
Presisi	92.86%

Recall	81.25%
F1-Score	86.67%

Confusion Matrix	Prediksi: Disorder	Prediksi: Normal
Aktual: Disorder	13	3
Aktual: Normal	1	14

c. RBF Kernel

Hasil Evaluasi:

Metrik	Nilai
Akurasi	87.10%
Presisi	87.50%
Recall	87.50%
F1-Score	87/50%

Confusion Matrix	Prediksi: Disorder	Prediksi: Normal
Aktual: Disorder	14	2
Aktual: Normal	2	13

Perbandingan Kernel

Kernel	Akurasi	Presisi	Reacall	F1-Score
Linier	83.87%	86.67%	81.25%	83.87%
Polynomial	87.10%	92,86%	81.25%	86.67%
RBF	87.10%	87.50%	87.50%	87.50%

Kesimpulan

Berdasarkan hasil evaluasi:

- Ketiga kernel memberikan hasil cukup tinggi, di atas 80% akurasi.
- Kernel RBF memberikan nilai terbaik secara seimbang pada semua metrik.
- Kernel Polynomial memiliki presisi tertinggi, menunjukkan kemampuan baik dalam mengenali kelas positif dengan benar.
- Kernel Linear sedikit lebih rendah dibanding dua lainnya, cocok untuk data yang linier separable.

Kesimpulan akhir: Kernel RBF merupakan pilihan terbaik untuk dataset ini karena menghasilkan metrik paling seimbang dan akurasi tertinggi.

Support Vector Machine Classifier

Upload your dataset, train multiple SVM kernels, and analyze the results

Step 1: Upload Dataset



Click to Upload CSV File

or drag and drop your file here

File uploaded successfully!

File: Sleep_health_and_lifestyle_dataset.csv | Rows: 374 | Columns: 13

Dataset Preview

Dataset Information: 374 rows × 13 columns

Person ID	Gender	Age	Occupation	Sleep Duration	Quality of Sleep	Physical Activity Level	Stress Level	BMI Category	Blood Pressure	Heart Rate	Daily Steps	Sleep Disorder
1	Male	27	Software Engineer	6.1	6	42	6	Overweight	126/83	77	4200	NaN
2	Male	28	Doctor	6.2	6	60	8	Normal	125/80	75	10000	NaN
3	Male	28	Doctor	6.2	6	60	8	Normal	125/80	75	10000	NaN
4	Male	28	Sales Representative	5.9	4	30	8	Obese	140/90	85	3000	Sleep Apnea
5	Male	28	Sales Representative	5.9	4	30	8	Obese	140/90	85	3000	Sleep Apnea
6	Male	28	Software	5.9	4	30	8	Obese	140/90	85	3000	Insomnia

Step 2: Configure Training

Target Column:

Sleep Disorder

Test Size:

0,2

Training: 80% | Testing: 20%

Train SVM Models

Dataset Split Information

155

TOTAL SAMPLES

124

TRAINING SAMPLES

31

TESTING SAMPLES

2

CLASSES

Training Data

Person ID	Gender	Age	Occupation	Sleep Duration	Quality of Sleep	Physical Activity Level	Stress Level	BMI Category	Blood Pressure	Heart Rate	Daily Steps	Sleep Disorder
310	Female	52	Accountant	6.6	7	45	7	Overweight	130/85	72	6000	NaN
236	Male	44	Salesperson	6.3	6	45	7	Overweight	130/85	72	6000	NaN
188	Male	43	Salesperson	6.3	6	45	7	Overweight	130/85	72	6000	NaN
231	Female	44	Teacher	6.6	7	45	4	Overweight	135/90	65	6000	NaN
246	Female	44	Teacher	6.5	7	45	4	Overweight	135/90	65	6000	NaN
223	Male	44	Salesperson	6.3	6	45	7	Overweight	130/85	72	6000	NaN
350	Female	57	Nurse	8.1	9	75	3	Overweight	140/95	68	7000	NaN

🏠 Training Data

Person ID	Gender	Age	Occupation	Sleep Duration	Quality of Sleep	Physical Activity Level	Stress Level	BMI Category	Blood Pressure	Heart Rate	Daily Steps	Sleep Disorder
310	Female	52	Accountant	6.6	7	45	7	Overweight	130/85	72	6000	NaN
236	Male	44	Salesperson	6.3	6	45	7	Overweight	130/85	72	6000	NaN
188	Male	43	Salesperson	6.3	6	45	7	Overweight	130/85	72	6000	NaN
231	Female	44	Teacher	6.6	7	45	4	Overweight	135/90	65	6000	NaN
246	Female	44	Teacher	6.5	7	45	4	Overweight	135/90	65	6000	NaN
223	Male	44	Salesperson	6.3	6	45	7	Overweight	130/85	72	6000	NaN
350	Female	57	Nurse	8.1	9	75	3	Overweight	140/95	68	7000	NaN
6	Male	28	Software Engineer	5.9	4	30	8	Obese	140/90	85	3000	Insomnia
239	Male	44	Salesperson	6.3	6	45	7	Overweight	130/85	72	6000	NaN
253	Female	45	Teacher	6.5	7	45	4	Overweight	135/90	65	6000	NaN
306	Female	51	Nurse	6.1	6	90	8	Overweight	140/95	75	10000	NaN
194	Male	43	Salesperson	6.5	6	45	7	Overweight	130/85	72	6000	NaN
187	Female	43	Teacher	6.7	7	45	4	Overweight	135/90	65	6000	NaN

🏠 Testing Data

Person ID	Gender	Age	Occupation	Sleep Duration	Quality of Sleep	Physical Activity Level	Stress Level	BMI Category	Blood Pressure	Heart Rate	Daily Steps	Sleep Disorder
354	Female	58	Nurse	8.0	9	75	3	Overweight	140/95	68	7000	NaN
255	Female	45	Teacher	6.5	7	45	4	Overweight	135/90	65	6000	NaN
345	Female	57	Nurse	8.2	9	75	3	Overweight	140/95	68	7000	NaN
196	Male	43	Salesperson	6.5	6	45	7	Overweight	130/85	72	6000	NaN
274	Female	49	Nurse	6.2	6	90	8	Overweight	140/95	75	10000	NaN
308	Female	52	Accountant	6.5	7	45	7	Overweight	130/85	72	6000	NaN
357	Female	58	Nurse	8.0	9	75	3	Overweight	140/95	68	7000	NaN
245	Male	44	Salesperson	6.3	6	45	7	Overweight	130/85	72	6000	NaN
199	Male	43	Salesperson	6.5	6	45	7	Overweight	130/85	72	6000	NaN
276	Female	49	Nurse	6.2	6	90	8	Overweight	140/95	75	10000	NaN
94	Male	35	Lawyer	7.4	7	60	5	Obese	135/88	84	3300	NaN
222	Male	44	Salesperson	6.4	6	45	7	Overweight	130/85	72	6000	NaN
244	Female	44	Teacher	6.5	7	45	4	Overweight	135/90	65	6000	NaN
4	Male	28	Software Engineer	5.9	4	30	8	Obese	140/90	85	3000	Insomnia

📊 LINEAR Kernel Results

83.87%

ACCURACY

86.67%

PRECISION

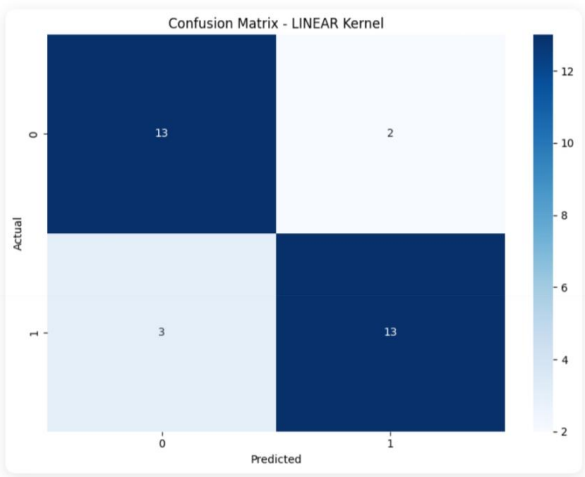
81.25%

RECALL

83.87%

F1-SCORE

📊 Confusion Matrix



📊 Manual Prediction

Person ID

Enter value

Gender

Female

Age

Enter value

Manual Prediction

Person ID

Enter value

Gender

Female

Age

Enter value

Occupation

Accountant

Sleep Duration

Enter value

Quality of Sleep

Enter value

Physical Activity Level

Enter value

Stress Level

Enter value

BMI Category

Normal

Blood Pressure

115/75

Heart Rate

Enter value

Daily Steps

Enter value

Predict (LINEAR)

Predicted Class:

POLYNOMIAL Kernel Results

87.10%

ACCURACY

92.86%

PRECISION

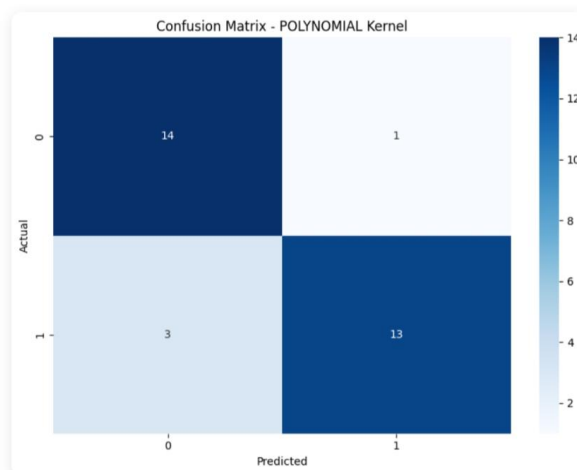
81.25%

RECALL

86.67%

F1-SCORE

Confusion Matrix



Manual Prediction

Person ID

Enter value

Gender

Female

Age

Enter value

Occupation

Accountant

Sleep Duration

Enter value

Quality of Sleep

Enter value

Physical Activity Level

Enter value

Stress Level

Enter value

BMI Category

Normal

Blood Pressure

115/75

Heart Rate

Enter value

Daily Steps

Enter value

Predict (POLYNOMIAL)

Predicted Class:

RBF Kernel Results

87.10%

ACCURACY

87.50%

PRECISION

87.50%

RECALL

87.50%

F1-SCORE

87.10%

ACCURACY

87.50%

PRECISION

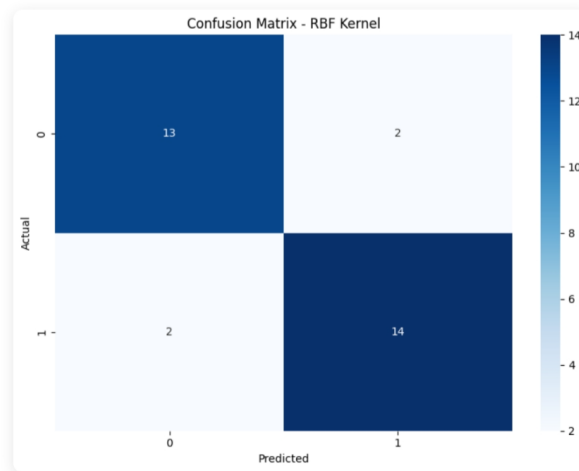
87.50%

RECALL

87.50%

F1-SCORE

Confusion Matrix



Manual Prediction

Person ID

Enter value

Gender

Female

Age

Enter value

Occupation

Accountant

Sleep Duration

Enter value

Quality of Sleep

Enter value

Physical Activity Level

Enter value

Stress Level

Enter value

BMI Category

Normal

Blood Pressure

115/75

Heart Rate

Enter value

Daily Steps

Enter value

Predict (RBF)

Predicted Class: